Designing and Implementing a project on

STUDENT MANAGEMENT SYSTEM

Submitted to the

Savitribai Phule Pune University

In partial fulfillment for the award of the Degree of

Bachelor of Engineering

in

Information Technology

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2024-25

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CERTIFICATE

This is to certify that the mini project report entitled "**Student Management System**" being submitted by **Shivangi Raj** and **Bhagyashri Tekale** (Roll no. I3243,I3251) is a record of bonafide work carried out by her under the supervision of **Prof. Deepa Mane** in partial fulfilment of the requirement for TE(Information Technology Engineering) 2019 course of Savitribai Phule Pune University,Pune in the academic year 2024-2025 .

Date: Place: Pune

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This Mini Project report has been examined by us as per the Savitribai Phule Pune University, Pune requirements at Smt.Kashibai Navale College of Engineering, Vadgaon,

Pune-41 on . . . .

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ACKNOWLEDGEMENT

We are very thankful to all the teachers who have provided us valuable guidance towards the completion of this project on **STUDENT MANAGEMENT SYSTEM** as part of the syllabus of bachelor's course. We express our sincere gratitude towards cooperative department who has provided us with valuable assistance and requirements for the system development.

We are very grateful and want to express our thanks to **Prof. D.B. Mane** for guiding us in right manner, correcting our doubts by giving us their time whenever we required, and providing their knowledge and experience in making this project. We are also thankful to the HOD of our Information Technology department **Dr.M.L.Bangare** for his moral support and motivation which has encouraged us in making this project. The acknowledgement will be incomplete if we do not thank our Principal **Prof. Dr. A. V. Deshpande,** who gave us his constant support and motivation which has been highly instrumental in making our project.

(Students Name & Signature)

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1. *ABSTRACT*

The Student Management System is a mini-project developed for efficient management of studentrelated information in an educational institution. The project aims to streamline various administrative tasks and enhance the overall student management process. It utilizes a relational database management system (DBMS) to store and manage student data, course details, attendance records, and academic performance.

The system provides a user-friendly interface for administrators and staff members to perform key functions, including student registration, course allocation, attendance tracking, and grade management. It offers robust features such as data validation, security measures, and data retrieval capabilities to ensure accurate and reliable information management.

The Student Management System is designed to address the following key objectives:

Student Registration: The system enables easy and efficient student registration by capturing essential details such as personal information, contact details, and previous educational records.

Course Management: Administrators can define and manage various courses offered by the institution, including course details, prerequisites, and credit hours.

Attendance Tracking: The system facilitates the recording and monitoring of student attendance for individual courses, providing insights into their overall attendance and punctuality.

Academic Performance: It allows staff members to input and manage student grades for different assignments, quizzes, and examinations. This feature provides a comprehensive overview of each student's academic performance.

Reporting and Analysis: The system generates comprehensive reports, including attendance reports, grade reports, and student profiles. These reports assist administrators in making informed decisions and analyzing student progress.

The Student Management System is an efficient tool for educational institutions to automate and streamline their student management processes. It enhances productivity, reduces manual paperwork, and improves data accuracy. By providing a centralized platform for managing student information, the system promotes effective communication and collaboration among stakeholders within the institution.

Keywords: Student Management System, DBMS, student registration, course management, attendance tracking, academic performance, reporting, educational institution.

2. INTRODUCTION

Student Management System deals with all kind of student details, academic related reports, college details, course details, curriculum, batch details and other resource related details too. It tracks all the details of a student from the day one to the end of his course which can be used for all reporting purpose, progress in the course, coming semester year curriculum details, project or any other assignment details, final exam result etc. Our design can facilitate us to explore all the activities happening in the college, even we can get to know which faculty is assigned to which course, the current status of a student, attendance percentage of a student and upcoming requirements of a student. The student management system is an automated version of manual Student Management System. It can handle all details about a student. The details include college details, subject details, student personnel details, academic details, exam details etc. In case of manual system they need a lot of time, manpower etc. Here almost all work is computerized. So the accuracy is maintained. Maintaining backup is very easy. It can do with in a few minutes. Our system has two type of accessing modes, administrator and user. Student management system is managed by an administrator. It is the job of the administrator to insert update and monitor the whole process. When a user log in to the system. He/she would only view details of the student. He/she can't perform any changes. Our system has seven modules, they are administrator, student, course, department, exam, attendance, and section. These modules and its attributes with entity relationship module presented in the ER diagram secion.

2.1 Problem Statement.

It was too difficult to make record of all students and faculty members manually, a lot of time and recourses were consumed to accomplish these above activities.

It was considering that a technology based project should be developed and implemented to carry out the above activities with minimum resources and time consumption,

So a web based project decided to be developed to manage above problems

2.2 Motivation

The motivation for using data modeling with the Entity-Relationship (ER) model in the Blood Bank Management System the use of data modeling with the ER model provides a structured and standardized approach to designing, organizing, and managing data within the Student Management System. It facilitates effective communication, supports database design and implementation, ensures data integrity, and enables future scalability and flexibility.

Advantages of DBMS over File Systems

Data redundancy and inconsistency –
Redundancy is the concept of repetition of data i.e. each data may have more than a single copy. The file system cannot control redundancy of data as each user defines and maintains the needed files for a specific application to run. Whereas DBMS controls redundancy by maintaining a single repository of data that is defined once and is accessed by many users. As there is no or less redundancy, data remains consistent.

Data sharing –
File system does not allow sharing of data or sharing is too complex. Whereas in DBMS, data can be shared easily due to centralized system.

Data searching –
For every search operation performed on file system, a different application program has to be written. While DBMS provides inbuilt searching operations.

2.3. Objectives

- The main objective of the project is to design and develop a user friendly-system Easy to use and an efficient computerized system.
- To develop an accurate and flexible system, it will eliminate data redundancy.
- To study the functioning of Students management System. To make a software fast in processing, with good user interface.

2. *Data Types*

1. Integer: one optional sign character (+ or -) followed by at least one digit (0-9). Leading and trailing blanks are ignored. No other character is allowed.
2. Varchar: It is used to store alpha numeric characters. In this data type we can set the maximum number of characters up to 8000 ranges by defaults SQL server will set the size to 50 characters range.
3. Date: The DATE data type accepts date values. No parameters are required when declaring a DATE data type. Date values should be specified in the form: YYYY-MM-DD. However, Point Base will also accept single digits entries for month and day values.
4. Time: The TIME data type accepts time values. No parameters are required when declaring a data type. Date values should be specified in the form: HH:MM: SS. An optional fractional value can be used to represent nanoseconds.

4. Data Modeling using ER Model:

4.1 REQUIREMENTS COLLECTION AND ANALYSIS

1. Introduction:

- Provide an overview of the student management system project.
- Identify the purpose and objectives of the system. - Describe the target users and stakeholders.

2. Functional Requirements:

- User Registration: Allow students, teachers, and administrators to register and create user accounts.
- User Login: Provide a secure login mechanism for authenticated access to the system.

- Student Profile: Enable students to create and manage their profiles, including personal information, contact details, and academic records.
- Course Management: Allow administrators to define and manage courses, including course details, prerequisites, and availability.
- Enrollment: Enable students to enroll in courses and manage their course schedule.
- Attendance Tracking: Provide a mechanism for teachers to track student attendance and generate attendance reports.
- Grading and Assessment: Allow teachers to enter grades and assessments for students, calculate GPA, and generate reports.
- Communication: Enable communication between students, teachers, and administrators through messaging or notifications.
- Reporting and Analytics: Provide comprehensive reports and analytics on student performance, attendance, and enrollment statistics.
- Timetable Management: Generate and manage class timetables for students and teachers.
- Data Security and Privacy: Implement appropriate security measures to protect student data and ensure compliance with privacy regulations.

3. Non-functional Requirements:

- Usability: The system should have an intuitive and user-friendly interface.
- Performance: The system should be able to handle a large number of users and data without significant delays.
- Reliability: The system should be available and responsive at all times, with minimal downtime. - Scalability: The system should be scalable to accommodate future growth and increased user load.
- Security: Implement robust security measures to protect student data from unauthorized access or breaches.
- Integration: The system should be able to integrate with other existing systems, such as student information systems or learning management systems.
- Accessibility: Ensure the system is accessible to users with disabilities, following accessibility standards. - Compatibility: The system should be compatible with multiple devices and browsers.

4. Constraints and Assumptions:

- Identify any constraints or limitations that may impact the project, such as budget or technology restrictions. - Document any assumptions made during the requirements gathering process.

5. Use Cases:

- Identify and describe various use cases to illustrate the system's functionality from different user perspectives (e.g., student, teacher, administrator).

6. Stakeholder Interviews and Feedback:
- Conduct interviews or gather feedback from key stakeholders (e.g., students, teachers, administrators) to validate and refine the requirements.

7. Documentation and Review:
- Document the collected requirements in a clear and concise manner. - Share the requirements document with stakeholders for review and approval.

8. Requirement Prioritization:
- Prioritize the requirements based on their importance, feasibility, and impact on project success.
- Identify any high-priority or must-have features that should be implemented first.

9. Change Management:
- Define a process for managing change requests and requirements modifications throughout the project lifecycle.

10. Requirement Traceability:
- Establish a mechanism to trace requirements throughout the development process, ensuring that they are properly implemented and tested.

4.1.1 Entity Types, Entity Sets, Attributes, and Keys

1. Student:

- Entity Set: All the students enrolled in the system.

- Attributes:

- Student ID (Key): A unique identifier for each student.

- Name: The student's full name.

- Date of Birth: The student's birth date.

- Gender: The student's gender.

- Contact Number: The student's phone number.
- Email Address: The student's email address.
- Address: The student's home address.
- Guardian Name: The name of the student's guardian. - Guardian Contact: The contact number of the student's guardian.

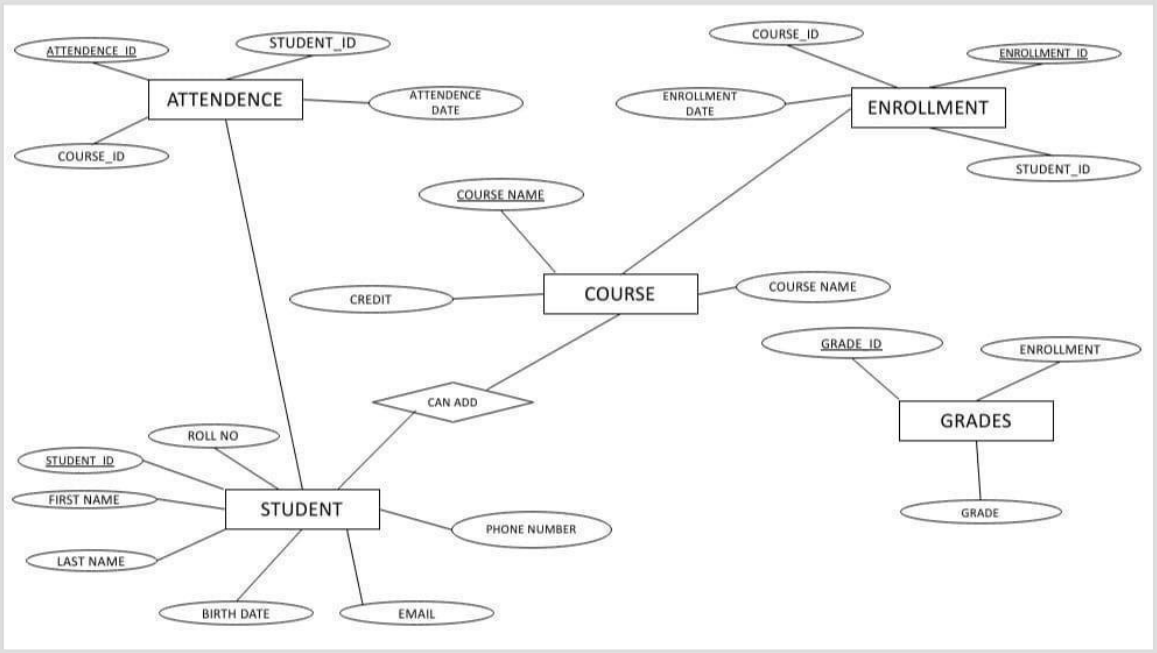
2. Course:
- Entity Set: All the courses available in the system.
- Attributes:
- Course ID (Key): A unique identifier for each course. - Course Name: The name of the course.
- Course Description: A brief description of the course.
- Credit Hours: The credit hours associated with the course. - Prerequisites: Any prerequisites required for the course.

3. Enrollment:
- Entity Set: All the enrollments made by students.
- Attributes:
- Enrollment ID (Key): A unique identifier for each enrollment.
- Student ID (Foreign Key): References the student who enrolled in the course.
- Course ID (Foreign Key): References the course in which the student is enrolled.
- Enrollment Date: The date when the student enrolled in the course. - Status: The status of the enrollment (e.g., enrolled, dropped, completed).

4. Attendance:
- Entity Set: All the attendance records for students in each course.
- Attributes:
- Attendance ID (Key): A unique identifier for each attendance record.
- Enrollment ID (Foreign Key): References the enrollment for which the attendance is recorded.
- Date: The date of the attendance record. - Status: The attendance status for the student on that date (e.g., present, absent, late).

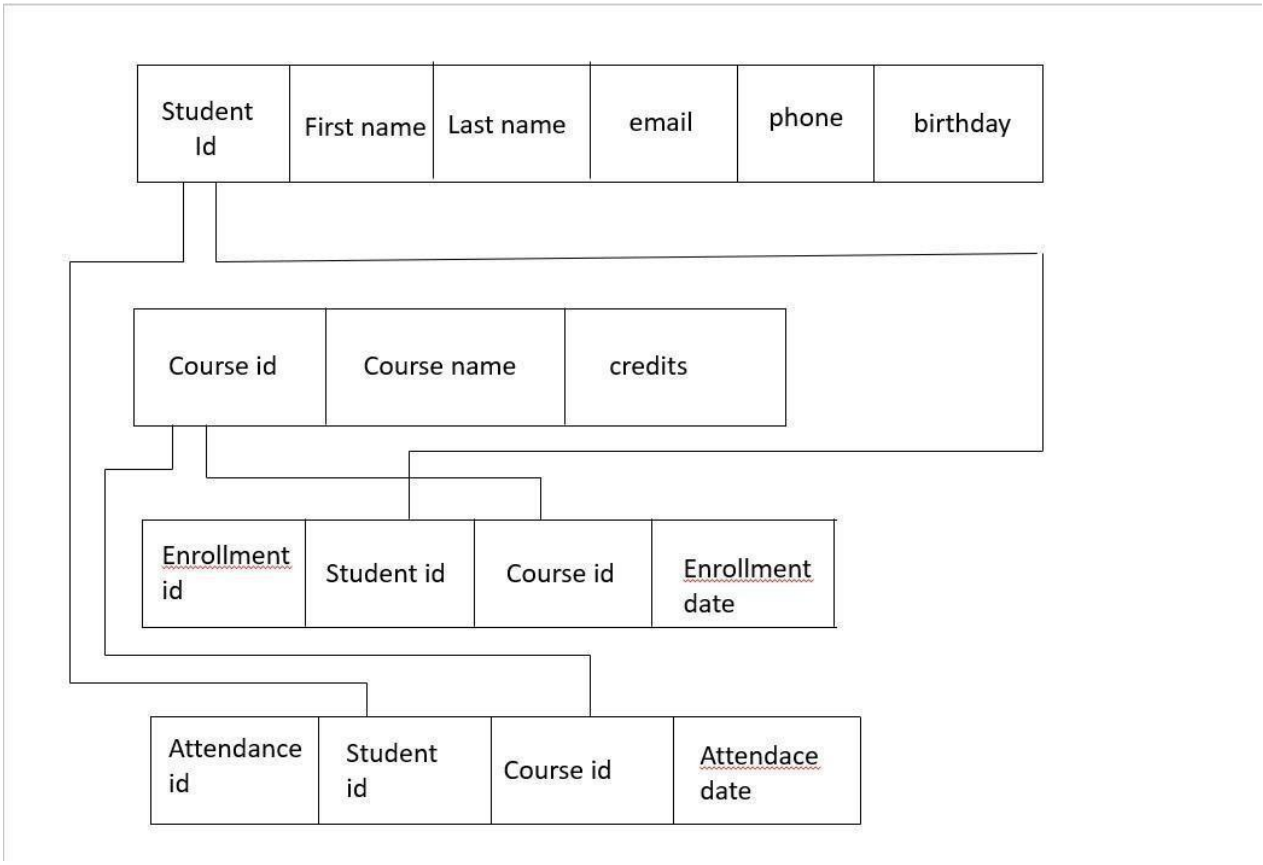
4.2 E-R Diagram

An entity–relationship model (ER model) describes inter-related things of interest in a specific domain of knowledge. An ER model is composed of entity types (which classify the things of interest) and specifies relationships that can exist between instances of those entity types. In software engineering an ER model is commonly formed to represent things that a business needs to remember in order to perform business processes. Consequently, the ER model becomes an abstract data model that defines a data or information structure that can be implemented in a database, typically a relational database. Entity–relationship modeling was developed for database design by Peter Chen and published in a 1976 paper. However, variants of the idea existed previously, some ER modelers show super and subtype entities connected by generalization specialization relationships, and an ER model can be used also in the specification of domain specific ontology. An ER model is typically implemented as a database. In a simple relational database implementation, each row of a table represents one instance of an entity type, and each field in a table represents an attribute type. In a relational database a relationship between entities is implemented by storing the primary key of one entity as a pointer or &quote; foreign key & quote; in the table of another entity There is a tradition for ER/data models to be built at two or three levels of abstraction. Note that the conceptual-logical-physical hierarchy below is used in other kinds of specification, and is different from the three-schema approach to software engineering.



The ER conceptual schema Student Management System

4.2.2 Relational Database Design Using ER-to-Relational Mapping



5. CREATING DATABASE USING MYSQL

An SQL schema is identified by a schema name, and includes an authorization identifier to indicate the user or account who owns the schema, as well as descriptors for each element in the schema. Schema elements include tables, constraints, views, domains, and other constructs (such as authorization grants) that describe the schema. A schema is created via the CREATE SCHEMA statement, which can include all the schema elements' definitions. Alternatively, the schema can be assigned a name and authorization identifier, and the elements can be defined later. The CREATE TABLE command is used to specify a new relation by giving it a name and specifying its attributes and initial constraints. The attributes are specified first, and each attribute is given a name, a data type to specify its domain of values, and any attribute constraints, such as NOT NULL. The key, entity integrity, and referential integrity constraints can be specified within the

CREATE TABLE statement after the attributes are declared, or they can be added later using the ALTER TABLE command (see Chapter 5). Figure 4.1 shows sample data definition statements in SQL for the student management relational database schema.

```
CREATE TABLE Students (
student_id INT PRIMARY KEY,
first_name VARCHAR(50),
last_name VARCHAR(50),   email
VARCHAR(100),   phone
VARCHAR(20),   birthdate DATE

-- Create Courses table CREATE
TABLE Courses (   course_id INT
PRIMARY KEY,   course_name
VARCHAR(100),   credits INT
);

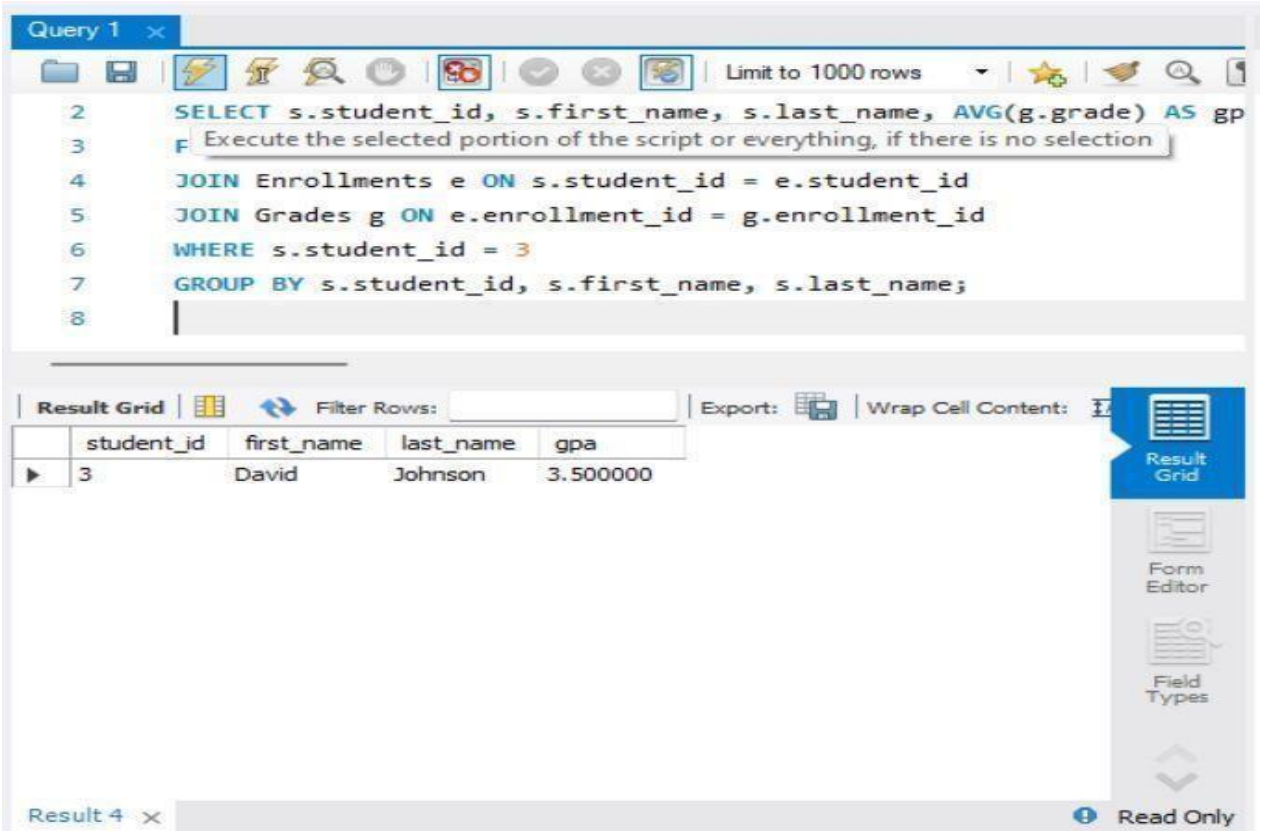
--   Create Enrollments table CREATE
TABLE Enrollments (   enrollment_id INT
PRIMARY KEY,   student_id INT,
 course_id INT,   enrollment_date
DATE,
  FOREIGN KEY (student_id) REFERENCES Students (student_id),
  FOREIGN KEY (course_id) REFERENCES Courses (course_id)
));                                                
```

```
  FOREIGN KEY (student_id) REFERENCES Students (student_id),
  FOREIGN KEY (course_id) REFERENCES Courses (course_id)
);
```

Snapshots

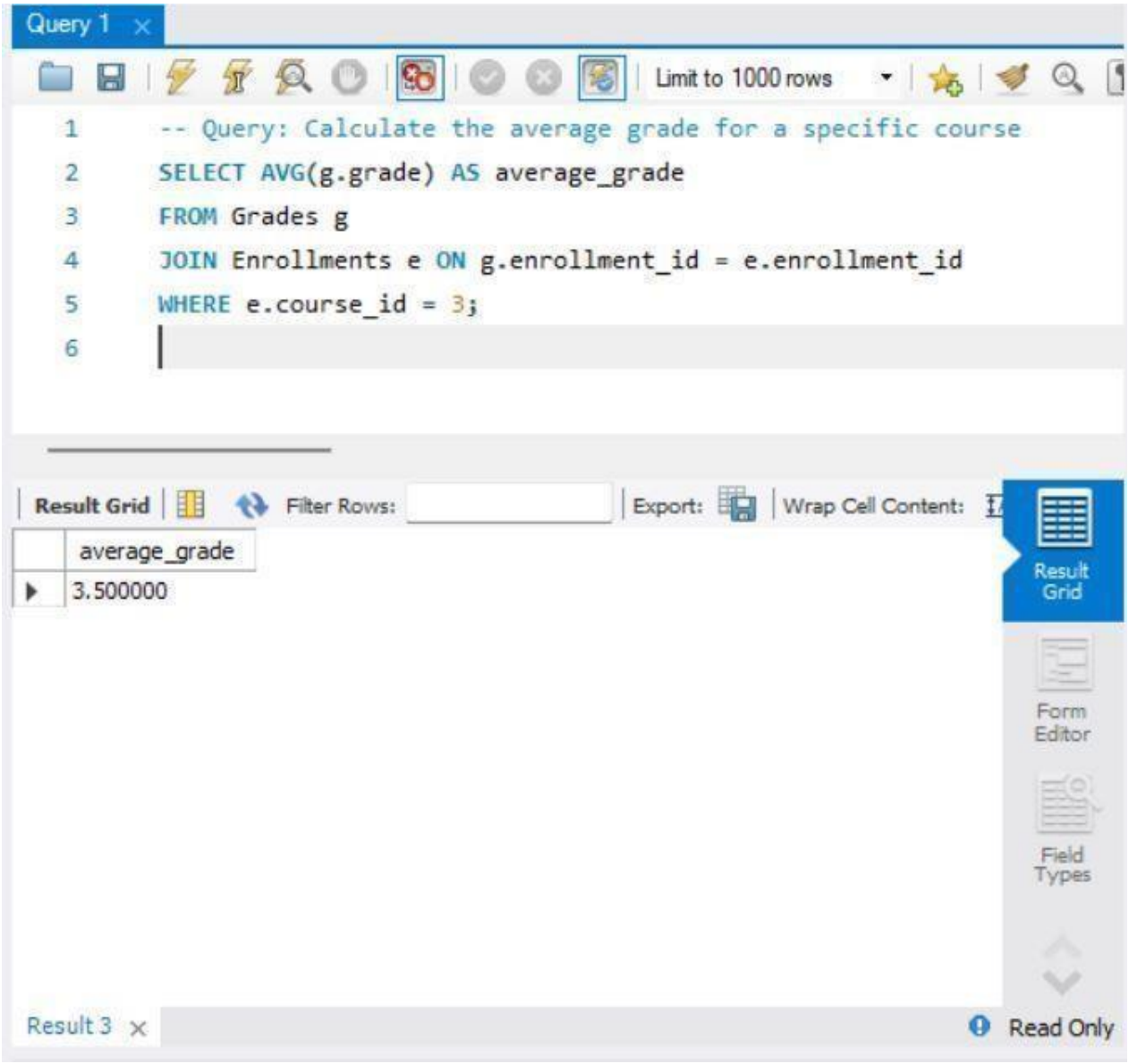
6. Test Queries

Query1-
Retrieve students
enrolled

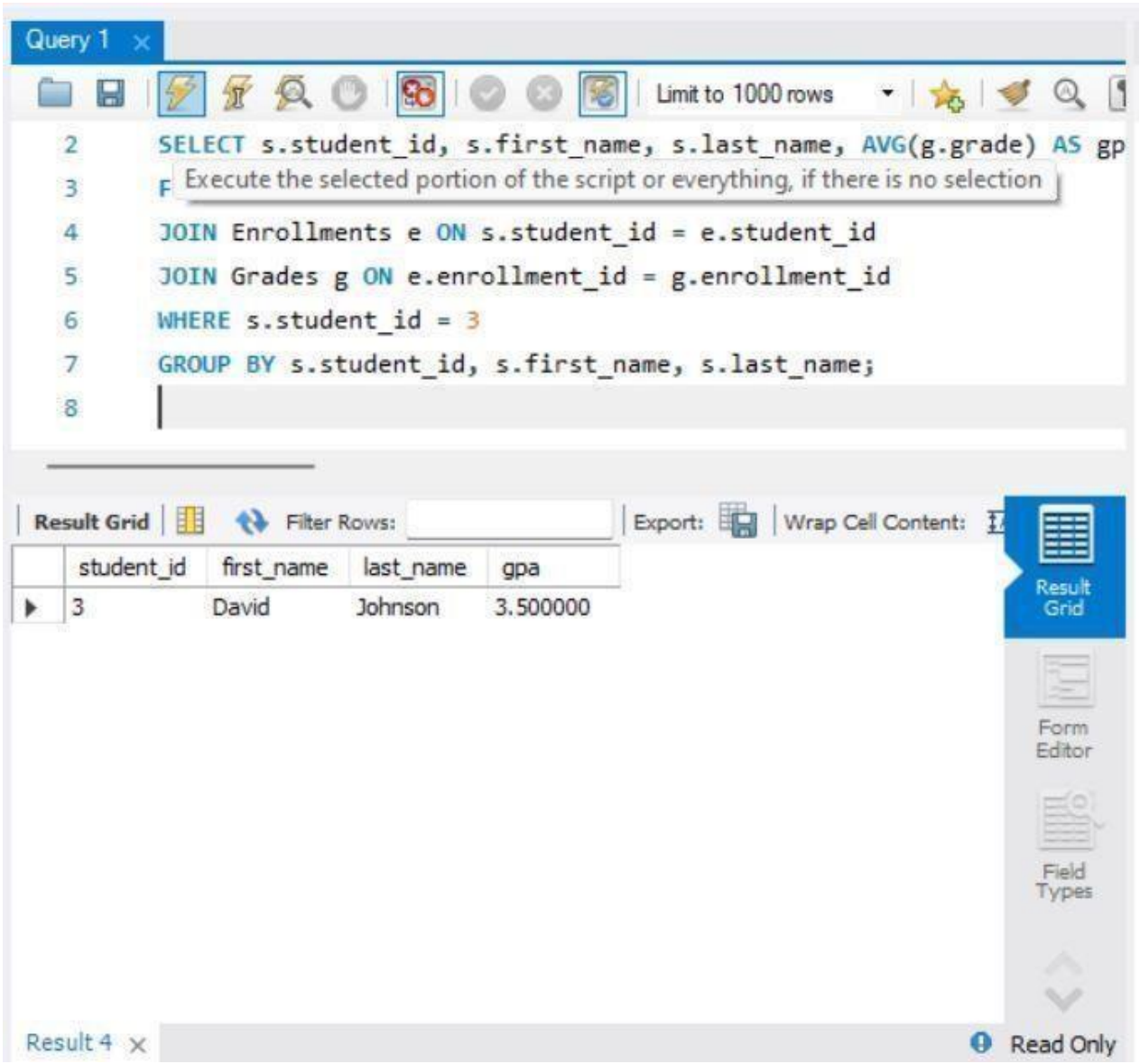


course

Query 2- Calculate the average grade for a specific course

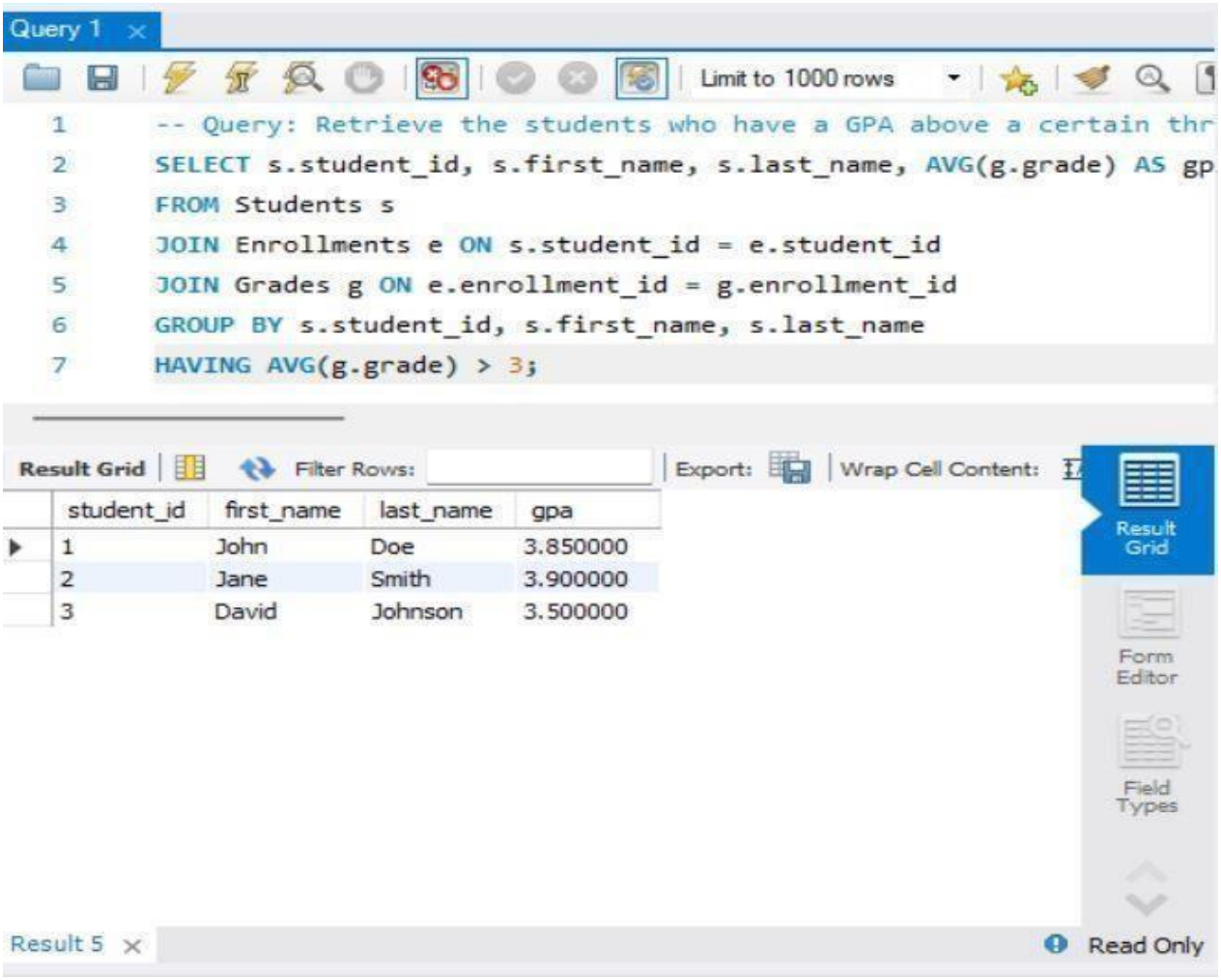


Query 3 Retrieve the overall performance or GPA for a students.

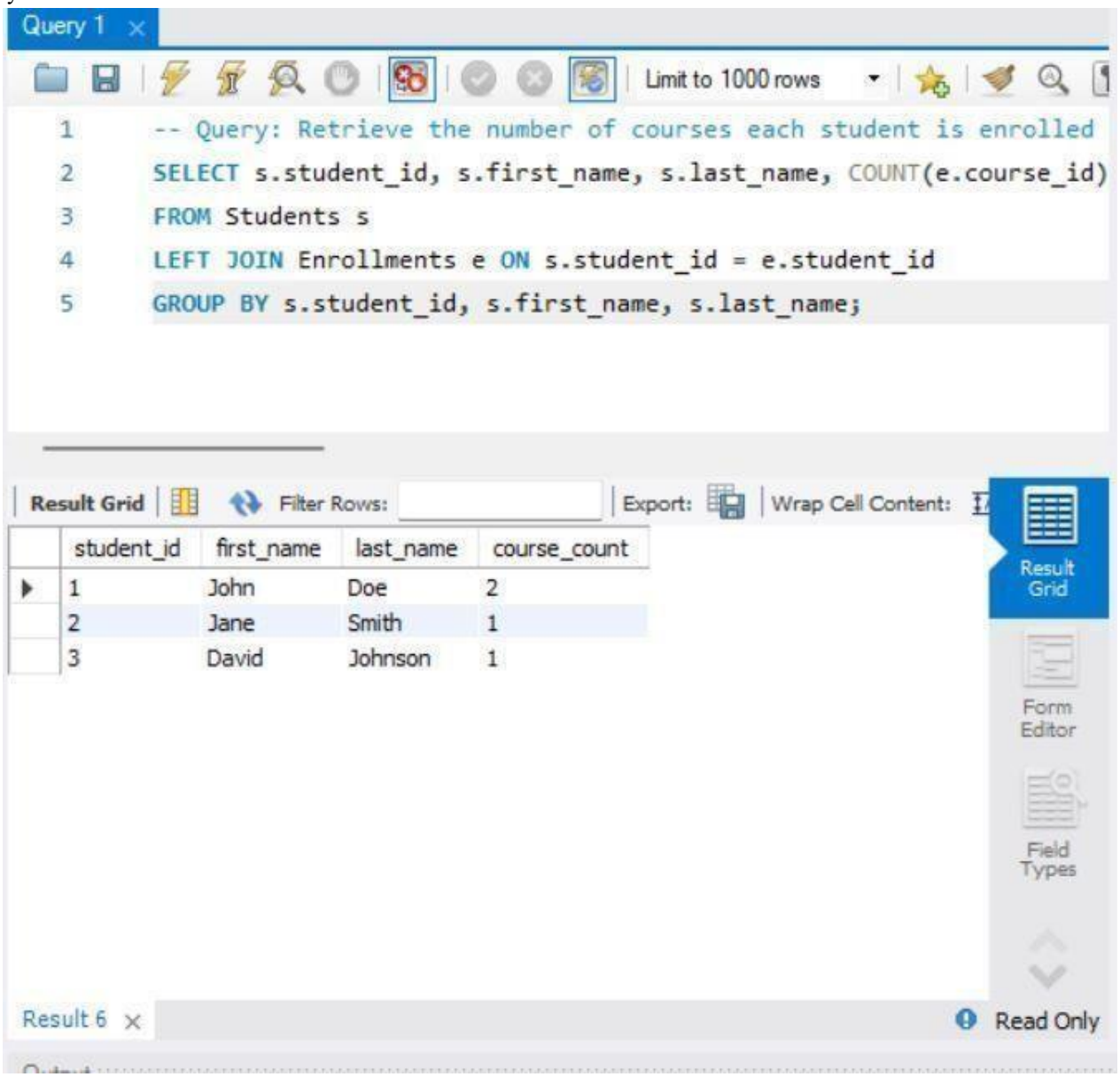


Query 4- Retrieve the students who

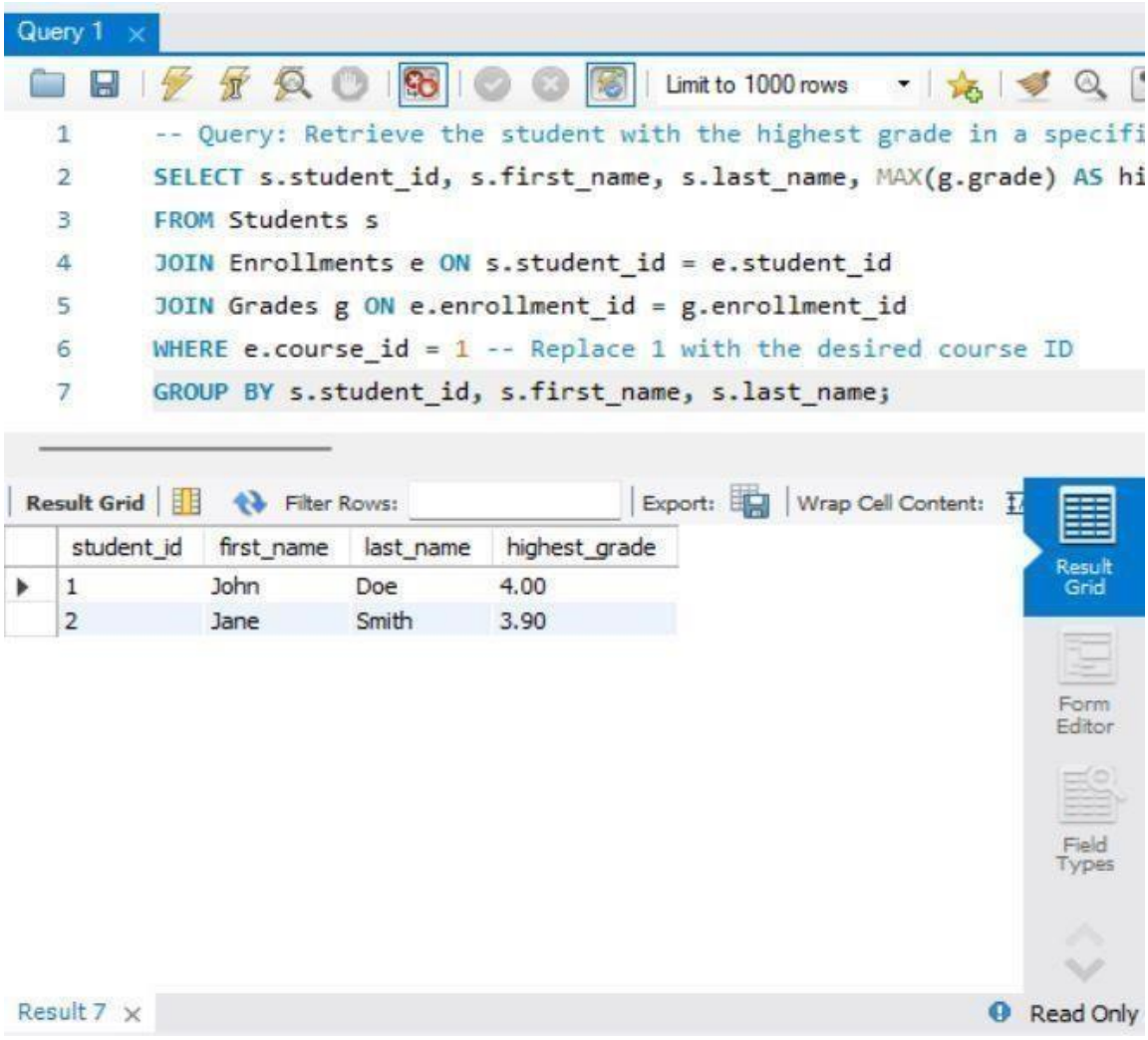
have GPA above
certain threshold.



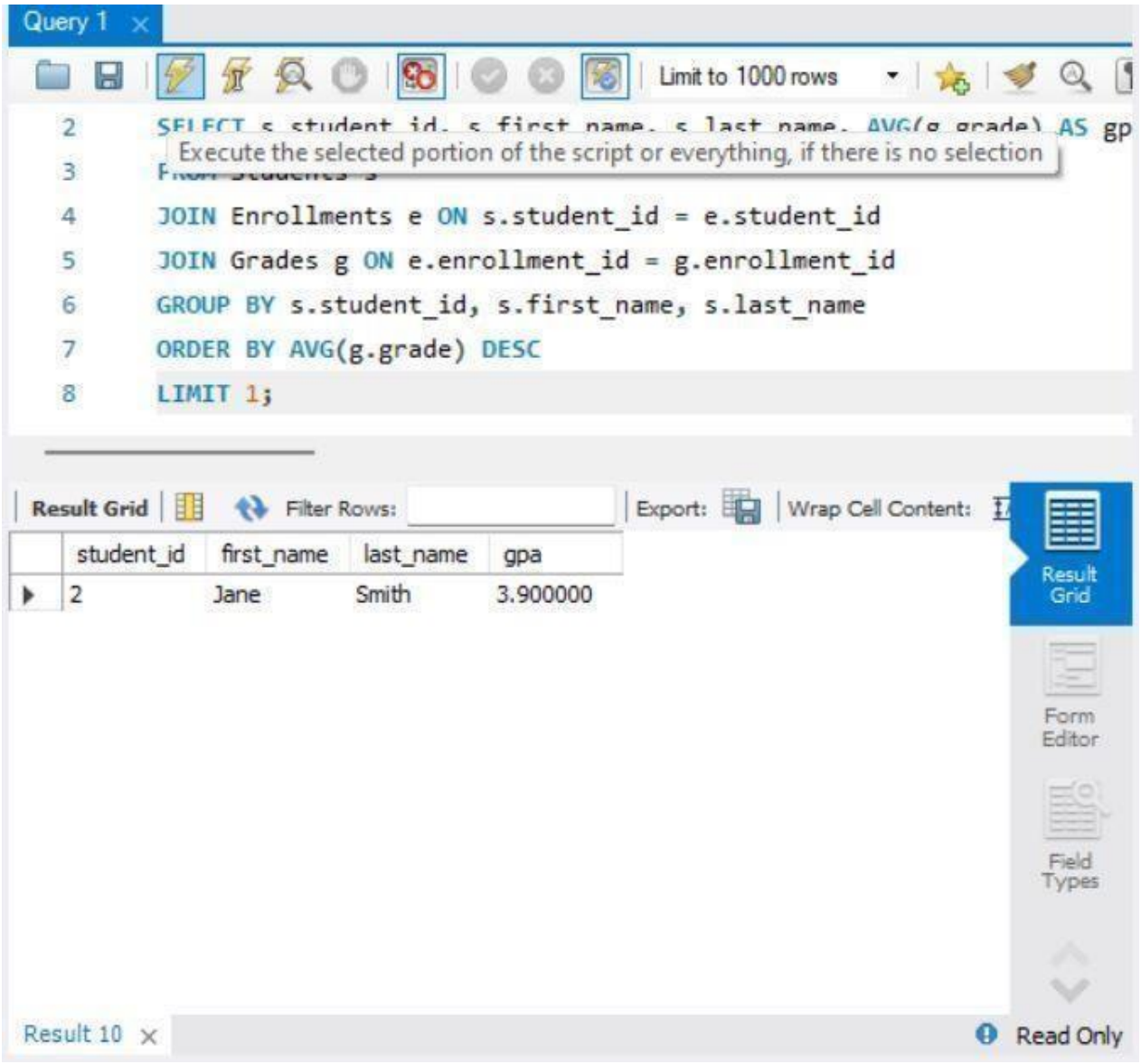
Query 5-Retrieve the number of courses each student is enrolled in



Query 6-
Retrieve the student with the highest grade in specific course.

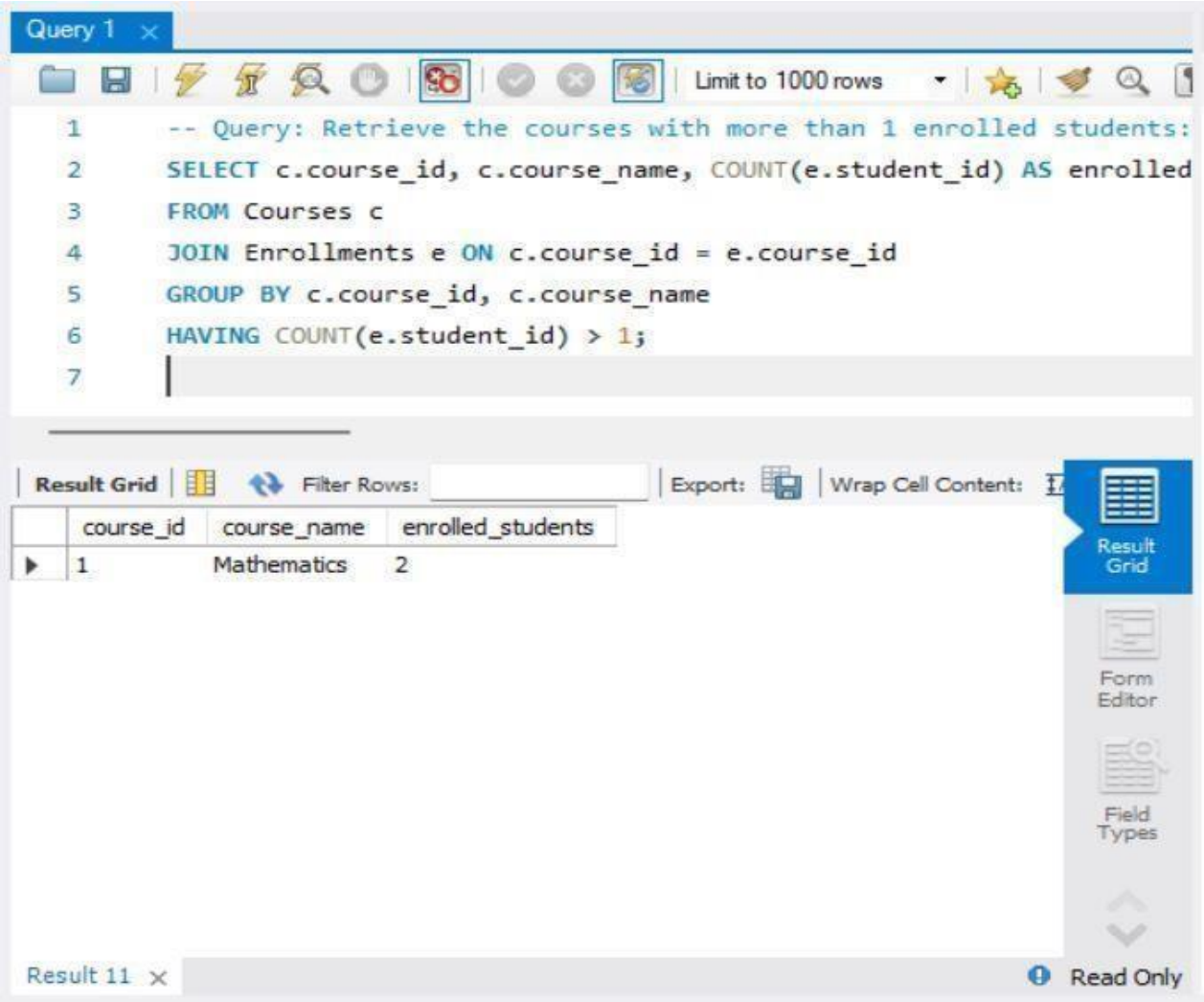


Query 7- Retrieve the students with the highest overall GPA:



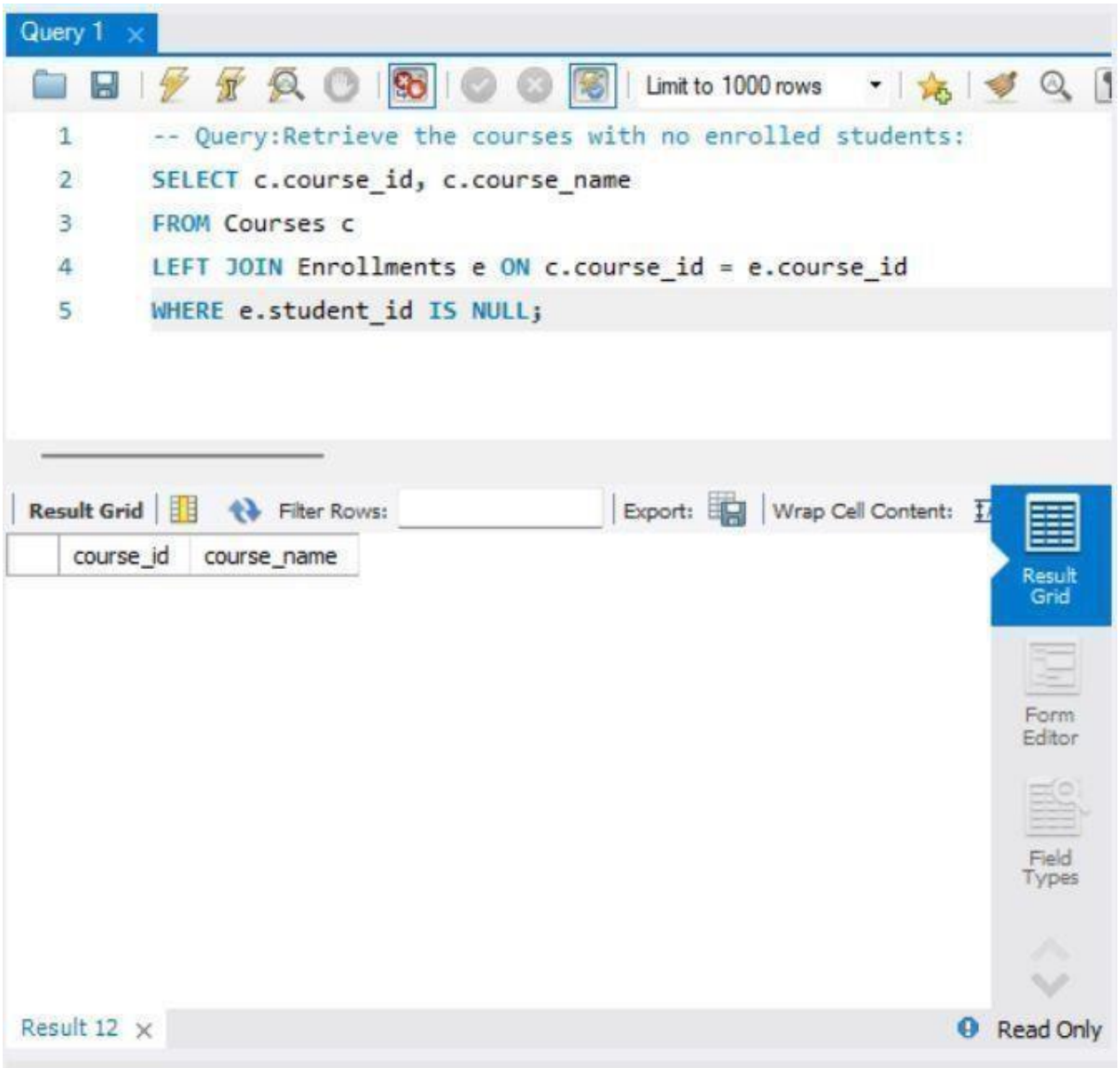
Query 8-

Retrieve the course with more than 1 enrolled students.



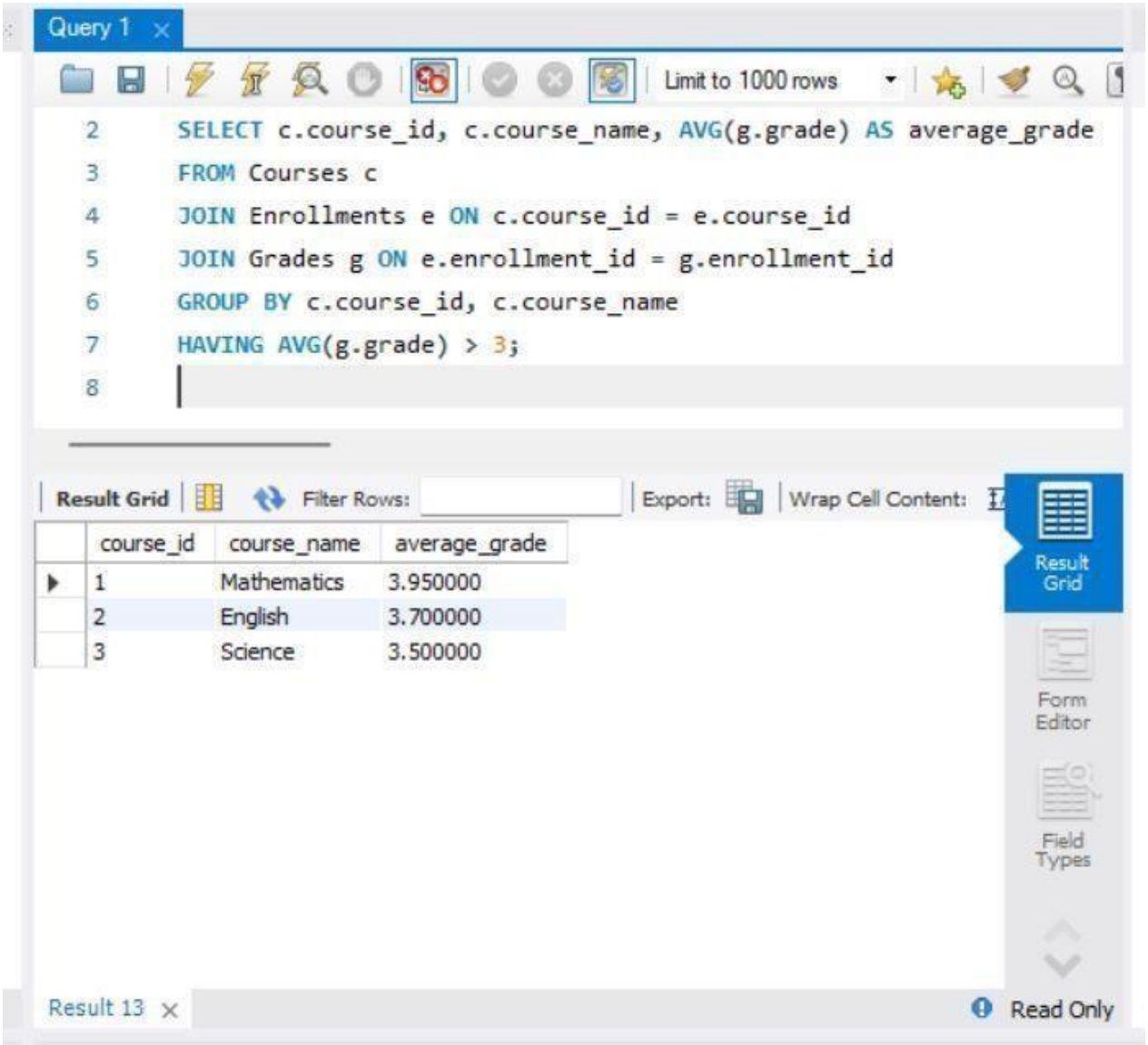
Query 9

Retrieve the courses with no enrolled students.



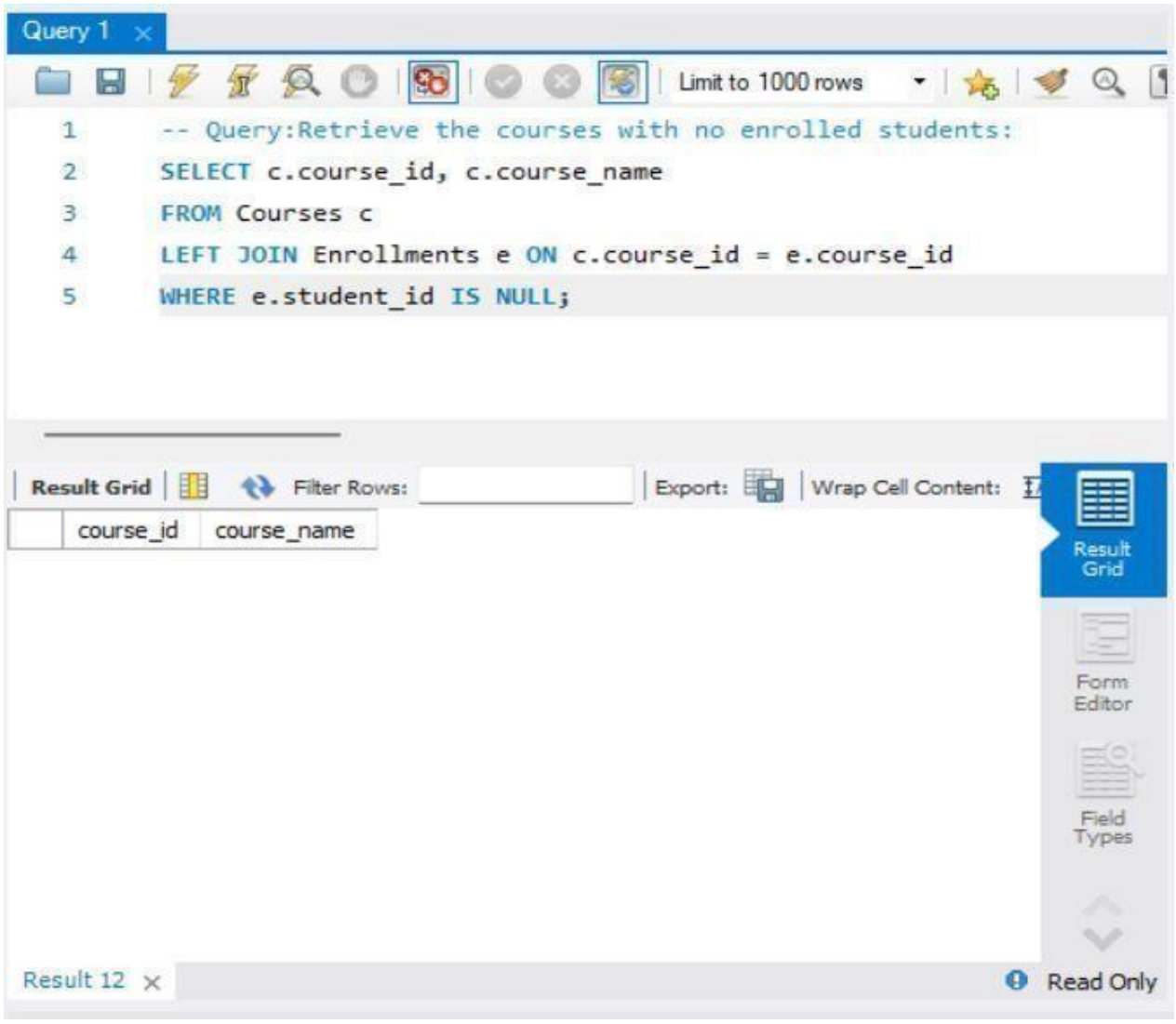
Query 10

Retrieve the courses where the average grade is above 3.



Query 11

Retrieve the students who are enrolled in all courses.



Query 12

7. Conclusion

STUDENT MANAGEMENT SYSTEM successfully implemented based on online data filling which helps us in administrating the data user for managing the tasks performed in students. The project successfully create the fully functional database management system for online portals.

Using MySQL as the database is highly beneficial as it is free to download, popular and can be easily customized. The data stored in the MySQL database can easily be retrieved and manipulated according to the requirements with basic knowledge of SQL.

With the theoretical inclination of our syllabus it becomes very essential to take the at most advantage of any opportunity of gaining practical experience that comes along. The building blocks of this Major Project “Students Management System” was one of these opportunities.

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